



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 30

Level 2 — Intermediate · Fractions I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Fractions I — Unlike Fractions & Fraction of a Quantity
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	30 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 30 takes fractions to Level 2: comparing and ordering **UNLIKE** fractions using a common denominator, finding a fraction OF a quantity ($\frac{3}{5}$ of 45), adding and subtracting simple unlike fractions ($\frac{1}{2} + \frac{1}{4}$), and two-step stories — pages left to read, fruits in a basket, money remaining. The 'half of a half' question plants the seed of fraction multiplication.

Learning Objectives

- Compare and order unlike fractions with a common denominator.
- Calculate a fraction of a quantity (divide, then multiply).
- Add and subtract simple unlike fractions (halves, thirds, quarters, sixths).
- Solve two-step fraction stories with money, pages and mixtures.
- Recognise improper fractions and quantities like $2\frac{1}{2}$ cups.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

Q1. Which is GREATER: $\frac{2}{3}$ or $\frac{3}{4}$?

- (A) $\frac{2}{3}$ (B) $\frac{3}{4}$
(C) Both equal (D) Cannot compare

Q2. What is $\frac{3}{5}$ of 45?

- (A) 9 (B) 18
(C) 27 (D) 36

Q3. What is $\frac{2}{7}$ of 63?

- (A) 9 (B) 14
(C) 18 (D) 21

Q4. Arrange in ASCENDING order: $\frac{1}{2}$, $\frac{2}{3}$, $\frac{1}{4}$

- (A) $\frac{1}{2}$, $\frac{2}{3}$, $\frac{1}{4}$ (B) $\frac{1}{4}$, $\frac{1}{2}$, $\frac{2}{3}$
(C) $\frac{2}{3}$, $\frac{1}{2}$, $\frac{1}{4}$ (D) $\frac{1}{4}$, $\frac{2}{3}$, $\frac{1}{2}$

Q5. A class has 40 students. $\frac{3}{8}$ of them are boys. How many GIRLS are there?

- (A) 15 (B) 20
(C) 25 (D) 30

Q6. What is $\frac{1}{2} + \frac{1}{4}$?

- (A) $\frac{2}{6}$ (B) $\frac{3}{4}$
(C) $\frac{1}{6}$ (D) $\frac{2}{4}$

Q7. What is $\frac{5}{6} - \frac{1}{3}$?

- (A) $\frac{4}{3}$ (B) $\frac{1}{2}$
(C) $\frac{2}{3}$ (D) $\frac{1}{6}$

Q8. Which fraction lies between $\frac{1}{4}$ and $\frac{1}{2}$?

- (A) $\frac{1}{8}$ (B) $\frac{3}{8}$
(C) $\frac{5}{8}$ (D) $\frac{7}{8}$

Q9. Ravi reads $\frac{2}{5}$ of a 200-page book. How many pages are LEFT to read?

- (A) 80 (B) 100
(C) 120 (D) 140

Q10. Which of these is an IMPROPER fraction (top $\geq$ bottom)?

- (A) $\frac{3}{4}$ (B) $\frac{7}{7}$
(C) $\frac{2}{9}$ (D) $\frac{5}{8}$

Q11. What is $\frac{3}{4}$ of one hour, in minutes?

- (A) 30 minutes (B) 40 minutes
(C) 45 minutes (D) 50 minutes



Q12. A recipe needs $\frac{1}{2}$ cup of sugar for one cake. How much sugar is needed for 5 cakes?

- (A) 2 cups (B) 2 and $\frac{1}{2}$ cups
(C) 3 cups (D) 1 and $\frac{1}{2}$ cups

Q13. Which is SMALLER: $\frac{4}{9}$ or $\frac{4}{7}$?

- (A) $\frac{4}{9}$ (B) $\frac{4}{7}$
(C) Both equal (D) Cannot compare

Q14. What fraction of **one rupee** is 25 paise?

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{1}{4}$ (D) $\frac{1}{5}$

Q15. A tank is $\frac{5}{8}$ full. What fraction MORE water will make it completely full?

- (A) $\frac{3}{8}$ (B) $\frac{5}{8}$
(C) $\frac{1}{8}$ (D) $\frac{2}{8}$

Q16. What is $\frac{1}{2}$ of $\frac{1}{2}$?

- (A) 1 (B) $\frac{1}{2}$
(C) $\frac{1}{4}$ (D) $\frac{1}{8}$

Q17. In a basket of 36 fruits, $\frac{1}{4}$ are apples, $\frac{1}{3}$ are bananas and the rest are oranges. How many ORANGES are there?

- (A) 12 (B) 15
(C) 9 (D) 21

Q18. Which is the GREATEST: $\frac{5}{6}$, $\frac{7}{12}$, $\frac{2}{3}$?

- (A) $\frac{5}{6}$ (B) $\frac{7}{12}$
(C) $\frac{2}{3}$ (D) All equal

Q19. Sham spent $\frac{1}{3}$ of his ₹90 pocket money. How much money does he have LEFT?

- (A) ₹30 (B) ₹45
(C) ₹60 (D) ₹75

Q20. A ribbon of $\frac{3}{4}$ m is cut from a 2 m ribbon. The remaining length is:

- (A) 1 and $\frac{1}{4}$ m (B) 1 and $\frac{1}{2}$ m
(C) 1 m (D) $\frac{3}{4}$ m

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	C	C	B	C	B	B	B	C	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	B	A	C	A	C	B	A	C	A

Step-by-Step Solutions

Q1. Correct answer: **(B)**

Common denominator 12: $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$. Since $9 > 8$, **$\frac{3}{4}$** is greater.

Q2. Correct answer: **(C)**

One-fifth of 45 = 9, so three-fifths = $3 \times 9 =$ **27**.

Q3. Correct answer: **(C)**

$63 \div 7 = 9$, then $\times 2 =$ **18**.

Q4. Correct answer: **(B)**

With denominator 12: $\frac{1}{4} = \frac{3}{12}$, $\frac{1}{2} = \frac{6}{12}$, $\frac{2}{3} = \frac{8}{12}$. Order: **$\frac{1}{4} < \frac{1}{2} < \frac{2}{3}$** .

Q5. Correct answer: **(C)**

Boys = $\frac{3}{8}$ of 40 = 15. Girls = $40 - 15 =$ **25**.

Q6. Correct answer: **(B)**

$\frac{1}{2} = \frac{2}{4}$. Then $\frac{2}{4} + \frac{1}{4} =$ **$\frac{3}{4}$** .

Q7. Correct answer: **(B)**

$\frac{1}{3} = \frac{2}{6}$. Then $\frac{5}{6} - \frac{2}{6} = \frac{3}{6} =$ **$\frac{1}{2}$** .

Q8. Correct answer: **(B)**

With denominator 8: $\frac{1}{4} = \frac{2}{8}$ and $\frac{1}{2} = \frac{4}{8}$. Between them is **$\frac{3}{8}$** .

Q9. Correct answer: **(C)**

Read = $\frac{2}{5}$ of 200 = 80 pages. Left = $200 - 80 =$ **120 pages**.

Q10. Correct answer: **(B)**

$\frac{7}{7}$ has numerator equal to denominator (it equals 1 whole), so it is improper. The others are proper fractions.

Q11. Correct answer: **(C)**

One hour = 60 minutes. $\frac{3}{4}$ of 60 = **45 minutes**.

Q12. Correct answer: **(B)**

$5 \times \frac{1}{2} = \frac{5}{2} =$ **2 and $\frac{1}{2}$ cups** (two halves make 1 cup, so 5 halves = $2\frac{1}{2}$).

Q13. Correct answer: **(A)**

Same numerator $\rightarrow$ the BIGGER denominator gives the smaller fraction: **$\frac{4}{9} < \frac{4}{7}$** .

Q14. Correct answer: **(C)**

₹1 = 100 paise. $\frac{25}{100} =$ **$\frac{1}{4}$** .

Q15. Correct answer: **(A)**

Full = $\frac{8}{8}$. Needed = $\frac{8}{8} - \frac{5}{8} =$ **$\frac{3}{8}$** .

**Q16. Correct answer: (C)**

Half of a half is a quarter: $\frac{1}{4}$. (Fold a paper in half, then in half again.)

Q17. Correct answer: (B)

Apples = 9, bananas = 12. Oranges = $36 - 9 - 12 = 15$.

Q18. Correct answer: (A)

Denominator 12: $\frac{5}{6} = \frac{10}{12}$, $\frac{7}{12} = \frac{7}{12}$, $\frac{2}{3} = \frac{8}{12}$. Greatest is $\frac{5}{6}$.

Q19. Correct answer: (C)

Spent = $90 \div 3 = ₹30$. Left = $90 - 30 = ₹60$.

Q20. Correct answer: (A)

$2 = \frac{8}{4}$. Remaining = $\frac{8}{4} - \frac{3}{4} = \frac{5}{4} = 1 \text{ and } \frac{1}{4} \text{ m.}$

